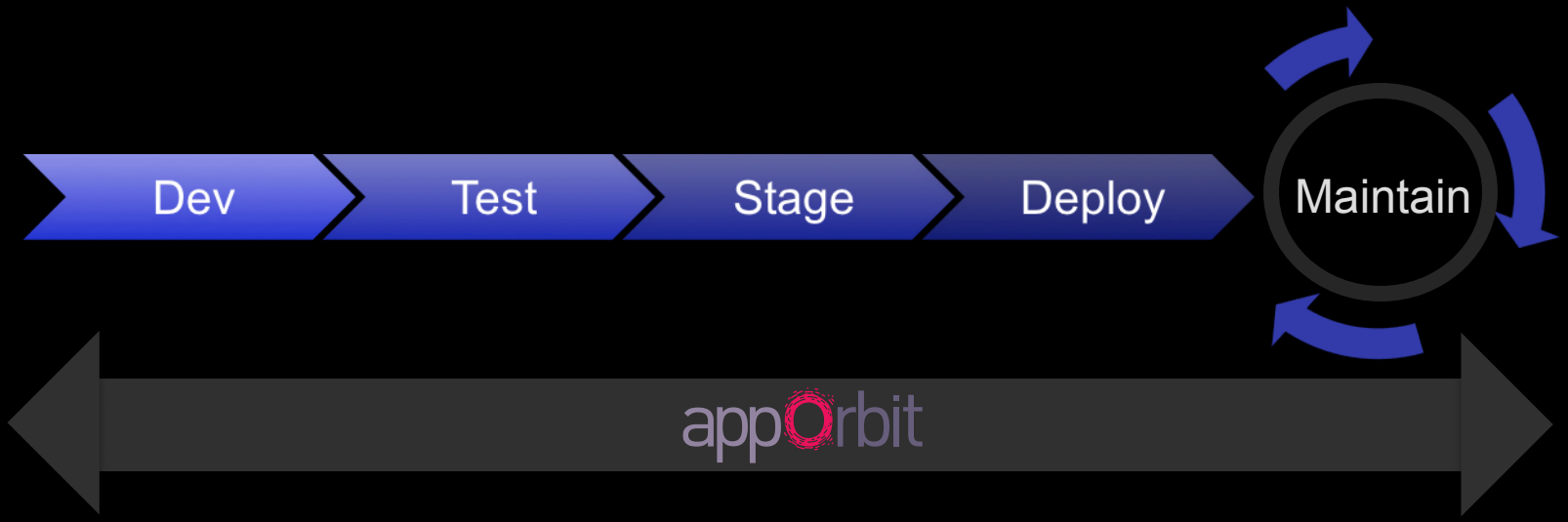
Record and vPlay
Debugging Container Crashes with "*Partial Checkpoints*"

Dinesh Subhraveti
CTO / Cofounder, appOrbit

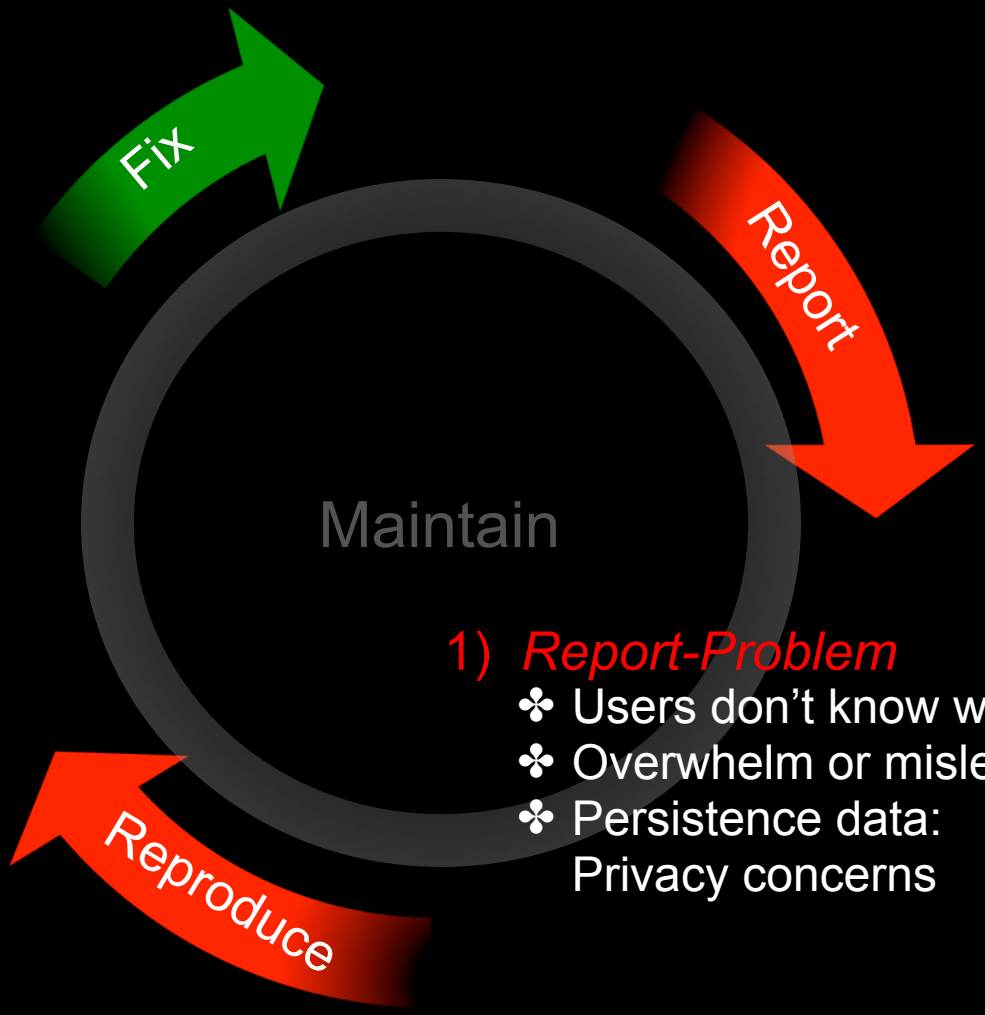
appOrbit

Application Lifecycle Management

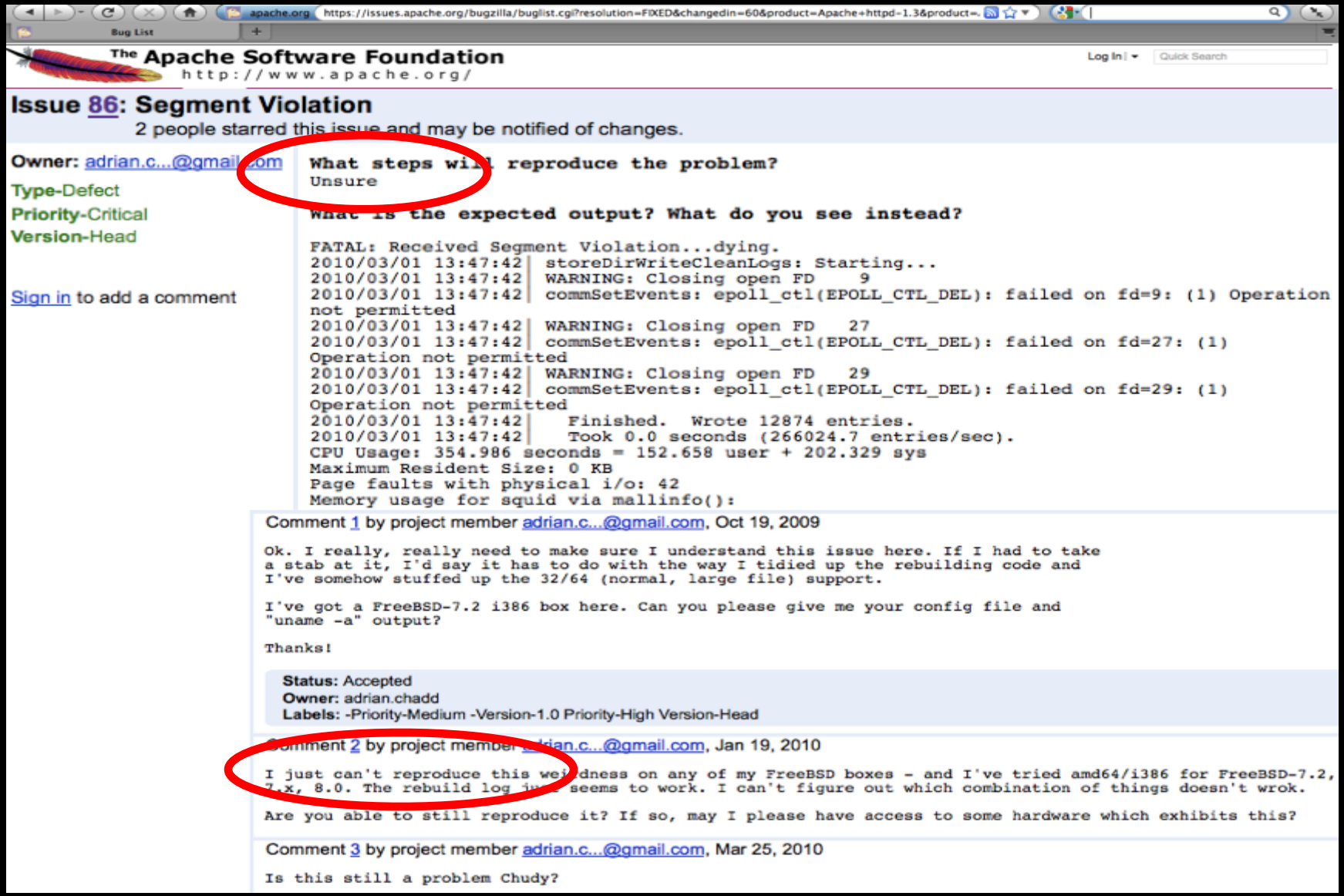


Key part of maintenance is rapid bug fixing

Problems with Problem Determination

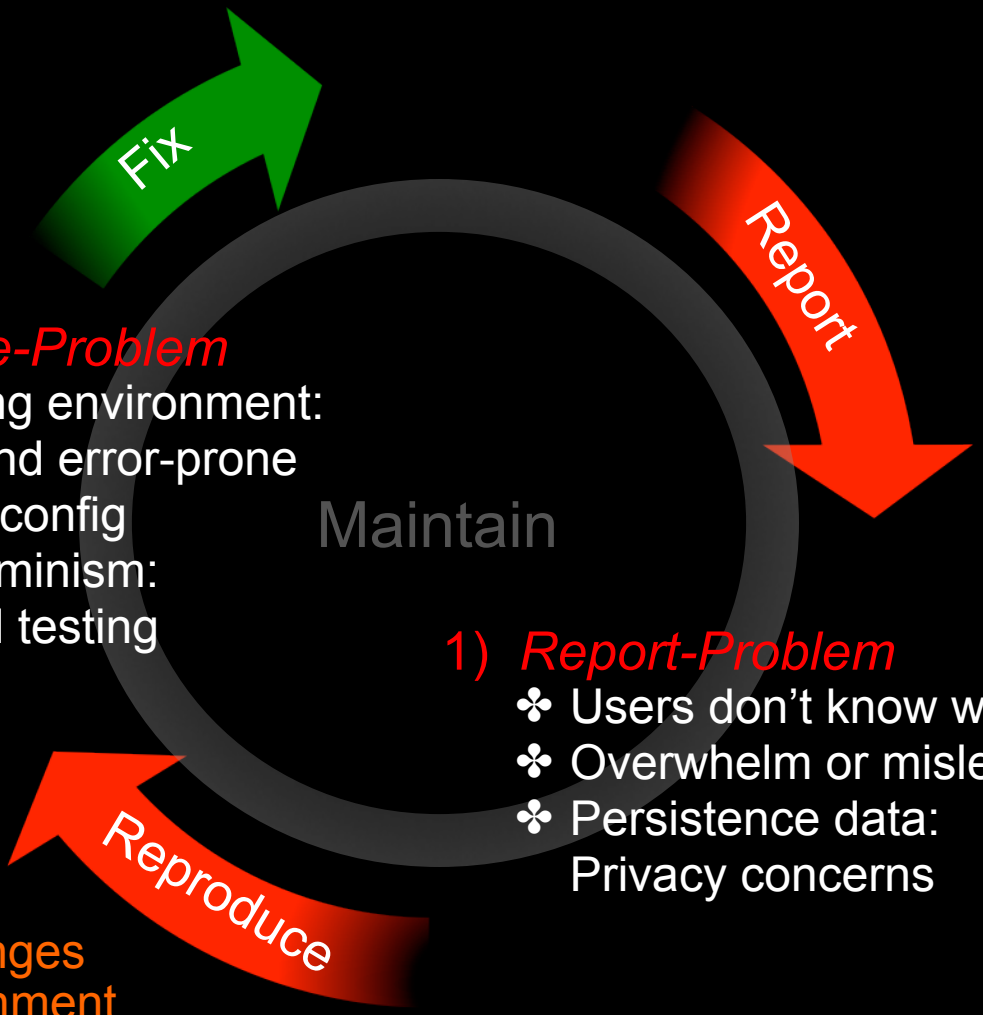


1) *Report-Problem*
   - ✤ Users don't know what is relevant
   - ✤ Overwhelm or mislead developer
   - ✤ Persistence data:
     Privacy concerns


apache.org https://issues.apache.org/bugzilla/buglist.cgi?resolution=FIXED&changedin=60&product=Apache+httpd-1.3&product=.
Bug List
The Apache Software Foundation
http://www.apache.org/
Log In
Quick Search
Issue 86: Segment Violation
2 people starred this issue and may be notified of changes.
Owner: adrian.c...@gmail.com
Type-Defect
Priority-Critical
Version-Head
Sign in to add a comment
What steps will reproduce the problem?
Unsure
What is the expected output? What do you see instead?
FATAL: Received Segment Violation...dying.
2010/03/01 13:47:42| storeDirWriteCleanLogs: Starting...
2010/03/01 13:47:42| WARNING: Closing open FD 9
2010/03/01 13:47:42| commSetEvents: epoll_ctl(EPOLL_CTL_DEL): failed on fd=9: (1) Operation not permitted
2010/03/01 13:47:42| WARNING: Closing open FD 27
2010/03/01 13:47:42| commSetEvents: epoll_ctl(EPOLL_CTL_DEL): failed on fd=27: (1) Operation not permitted
2010/03/01 13:47:42| WARNING: Closing open FD 29
2010/03/01 13:47:42| commSetEvents: epoll_ctl(EPOLL_CTL_DEL): failed on fd=29: (1) Operation not permitted
2010/03/01 13:47:42| Finished. Wrote 12874 entries.
2010/03/01 13:47:42| Took 0.0 seconds (266024.7 entries/sec).
CPU Usage: 354.986 seconds = 152.658 user + 202.329 sys
Maximum Resident Size: 0 KB
Page faults with physical i/o: 42
Memory usage for squid via mallinfo():
Comment 1 by project member adrian.c...@gmail.com, Oct 19, 2009
Ok. I really, really need to make sure I understand this issue here. If I had to take a stab at it, I'd say it has to do with the way I tidied up the rebuilding code and I've somehow stuffed up the 32/64 (normal, large file) support.
I've got a FreeBSD-7.2 i386 box here. Can you please give me your config file and "uname -a" output?
Thanks!
Status: Accepted
Owner: adrian.chadd
Labels: -Priority-Medium -Version-1.0 Priority-High Version-Head
Comment 2 by project member adrian.c...@gmail.com, Jan 19, 2010
I just can't reproduce this weirdness on any of my FreeBSD boxes - and I've tried amd64/i386 for FreeBSD-7.2, 7.x, 8.0. The rebuild log just seems to work. I can't figure out which combination of things doesn't wrok.
Are you able to still reproduce it? If so, may I please have access to some hardware which exhibits this?
Comment 3 by project member adrian.c...@gmail.com, Mar 25, 2010
Is this still a problem Chudy?

apporbit

Problems with Problem Determination

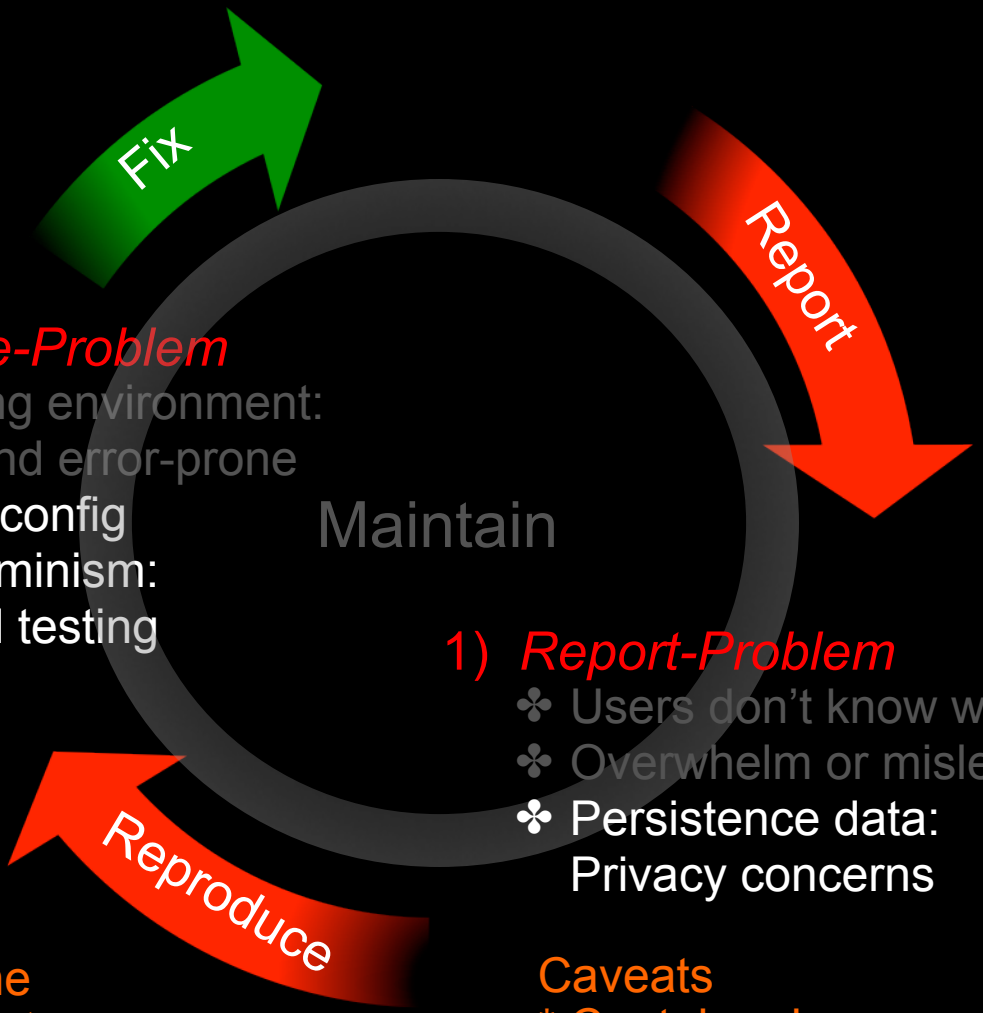


2) *Reproduce-Problem*
- ✤ Replicating environment: tedious and error-prone
- ✤ Complex config
- ✤ Nondeterminism: Repeated testing

1) *Report-Problem*
- ✤ Users don't know what is relevant
- ✤ Overwhelm or mislead developer
- ✤ Persistence data: Privacy concerns

Time spent on exchanges and setting up environment rather than fixing the problem

Problems with Problem Determination



2) *Reproduce-Problem*
- ✤ Replicating environment: tedious and error-prone
- ✤ Complex config
- ✤ Nondeterminism: Repeated testing

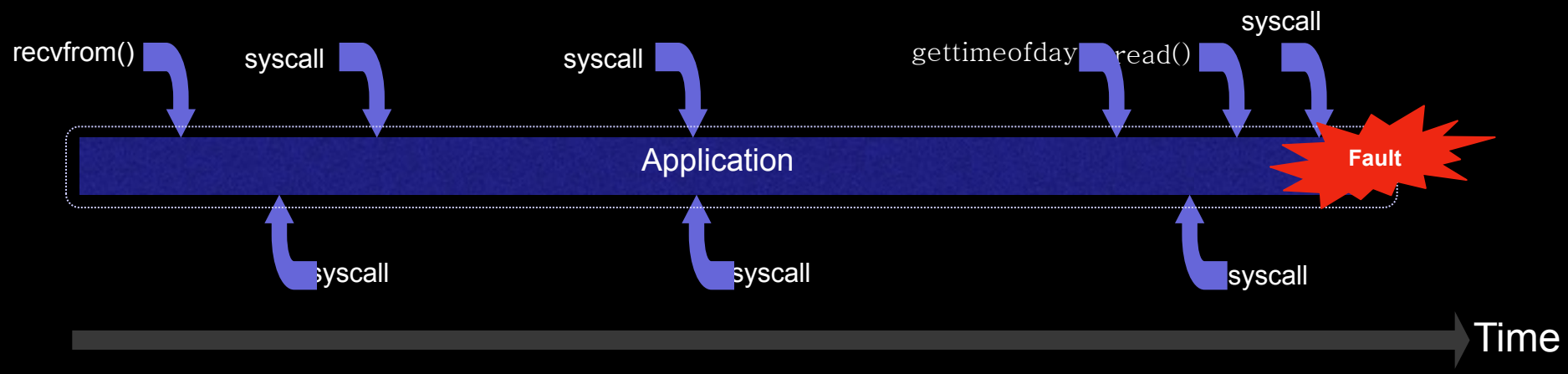
1) *Report-Problem*
- ✤ Users don't know what is relevant
- ✤ Overwhelm or mislead developer
- ✤ Persistence data: Privacy concerns

Containers address the problem to some extent

Caveats
* Container images are quite heavy
* Not cross-platform

Complete and Concise Recording

Application consumes a variety of data as inputs while executing

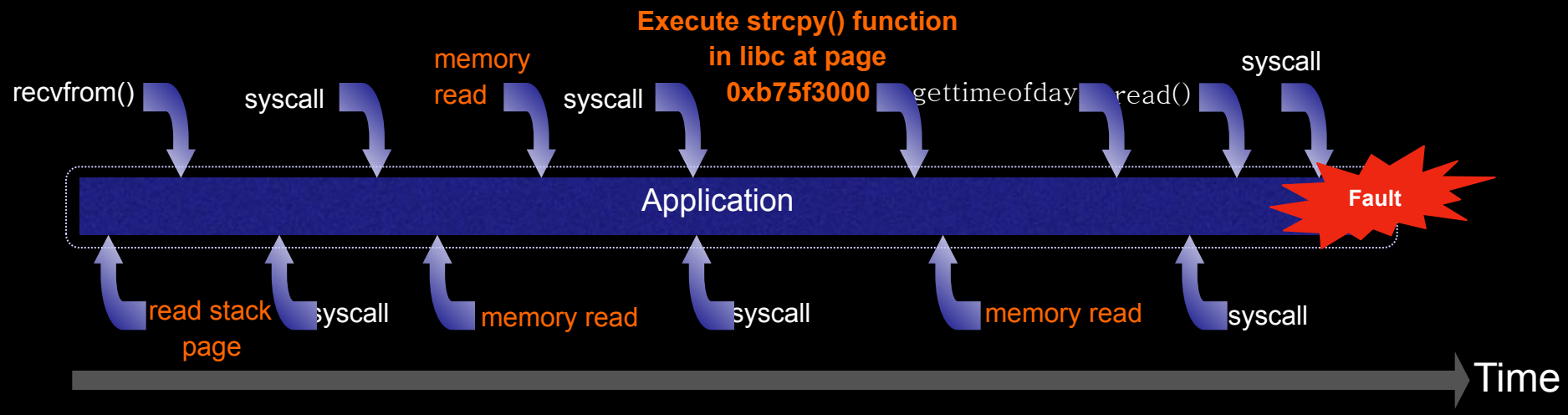


Inputs include

- Data read from files, network sockets etc.
- Data returned by the OS via system calls

Complete and Concise Recording

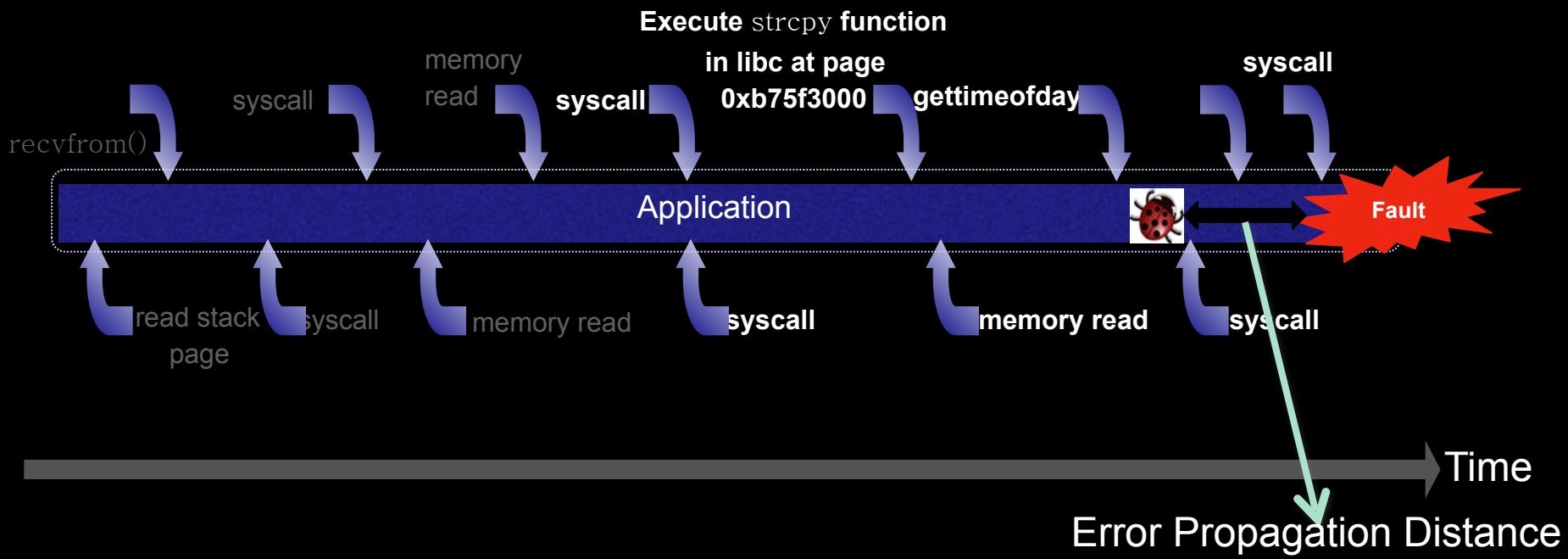
Application consumes a variety of data as inputs while executing



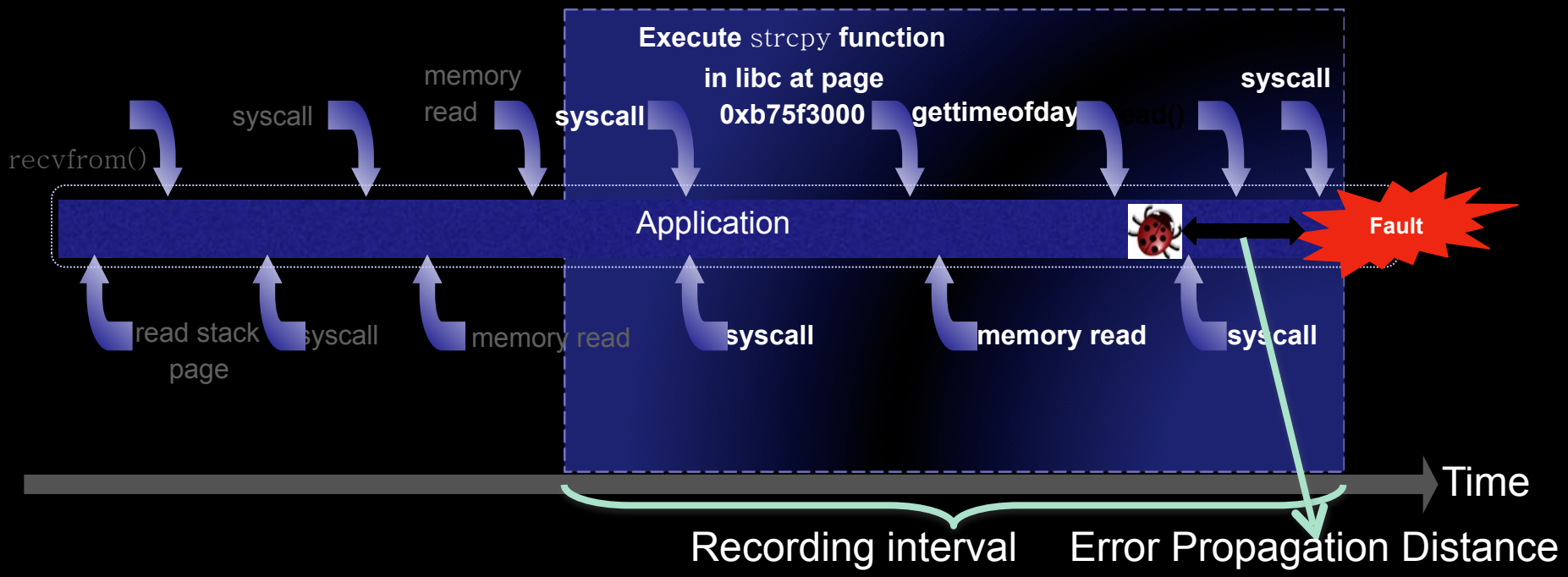
Inputs include

- Data read from files, network sockets etc.
- Data returned by the OS via system calls
- Bits of application and library code accessed by the application
- Memory pages read by the application

Partial Checkpointing



Partial Checkpointing



- Checkpoint: *Complete intermediate state of a running application at one point of its execution*
- Partial checkpoint: *Partial state of itself that an application accesses in a specified interval*

State Composition

- Processor context: At the beginning of the recording interval
- System calls: Results of system calls made by the application
- Virtual Memory: Memory pages accessed by the application
- Nondeterministic events: Meta data necessary for deterministic replay: interleaved shared memory accesses, signals

No Kernel state

Tracking Memory Pages

- Complications due to threads and changes in memory geometry
  - Processes and threads are created and deleted
  - Memory regions are added and removed: mappings change over time
  - Shared memory regions persist even without being attached to a process

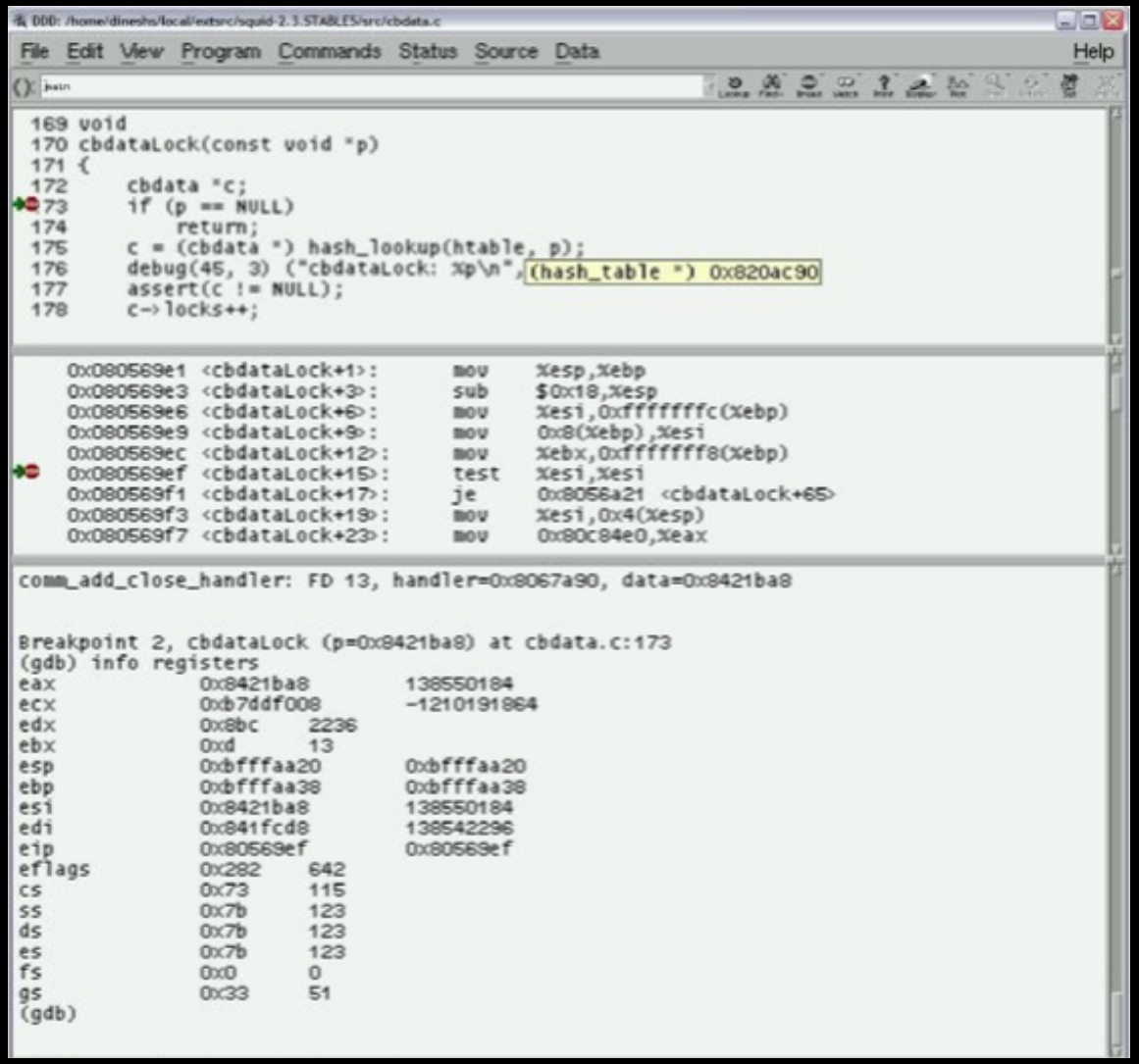
Virtual Replay

- Consists of two steps: Load phase and Replay phase
- Load Phase: performed by a purpose-built binary loader
  1) Sets up the initial sparsely-populated application address space
  2) Recursively creates application threads
  3) Transfers control to the application code as per register context

Replay Phase

- Execute the instructions produced by the application
- Most instructions are executed natively
  - **No privileged instructions**
- Two types of instructions need emulation
  - Instructions referencing user-defined segment registers (`fs, gs – modify_ldt()`)
  - Instructions that invoke a system call (`int 0x80, sysenter`)

Partial Checkpoint in a Debugger



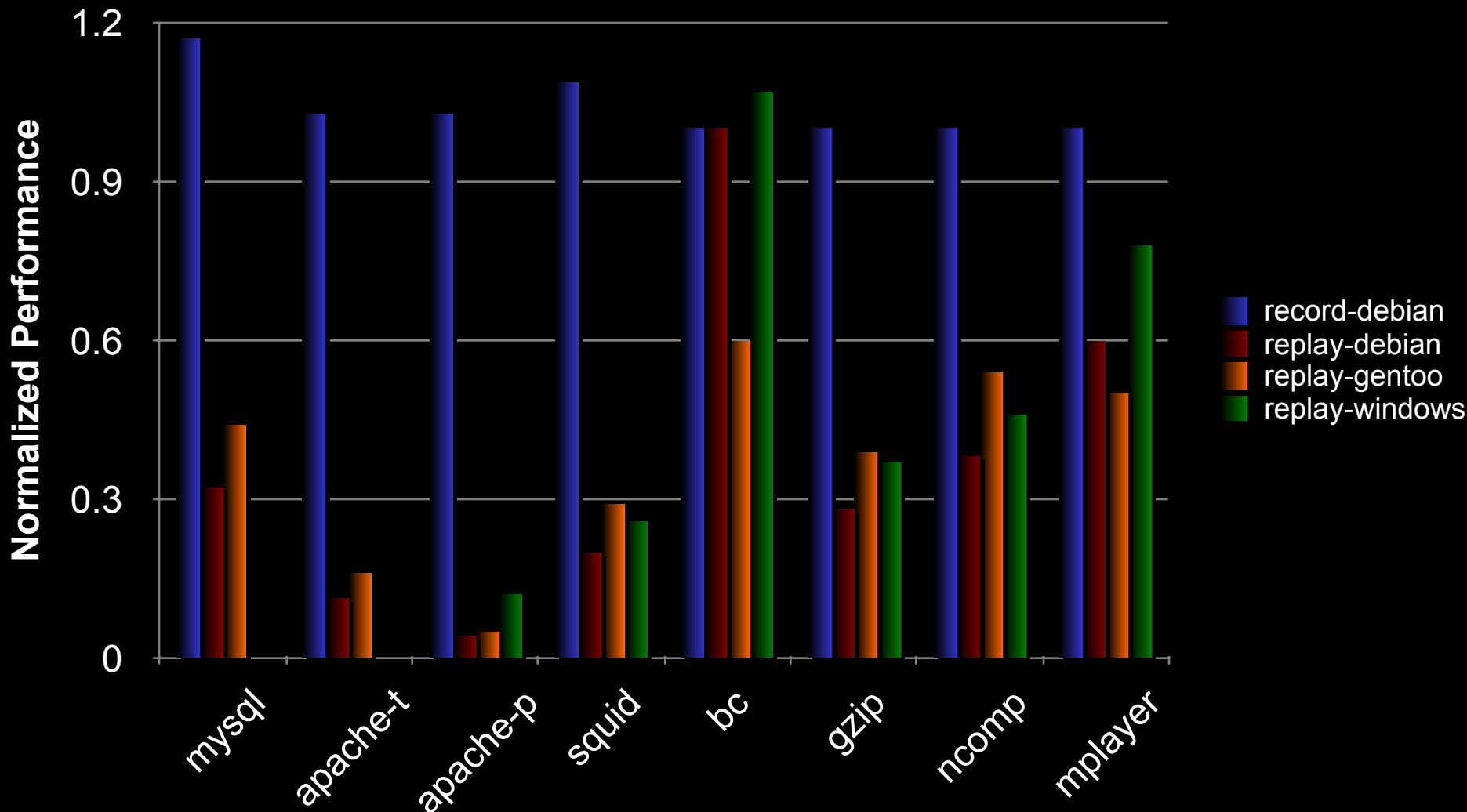
No different than debugging with a standard binary

Evaluation

Application	Workload	Bug
mysql	httperf (20,000 connection requests)	data race
apache	httperf (20,000 connection requests)	library incompatibility
squid	httperf (20,000 connection requests)	heap overflow
bc	Compute pi to 5000 decimal places	heap overflow
gzip	Compress 200 MB /dev/urandom data	global buffer overflow
ncomp	Compress 200 MB /dev/urandom data	stack smash
mplayer	Play 10 MB 1080p video file	device incompatibility

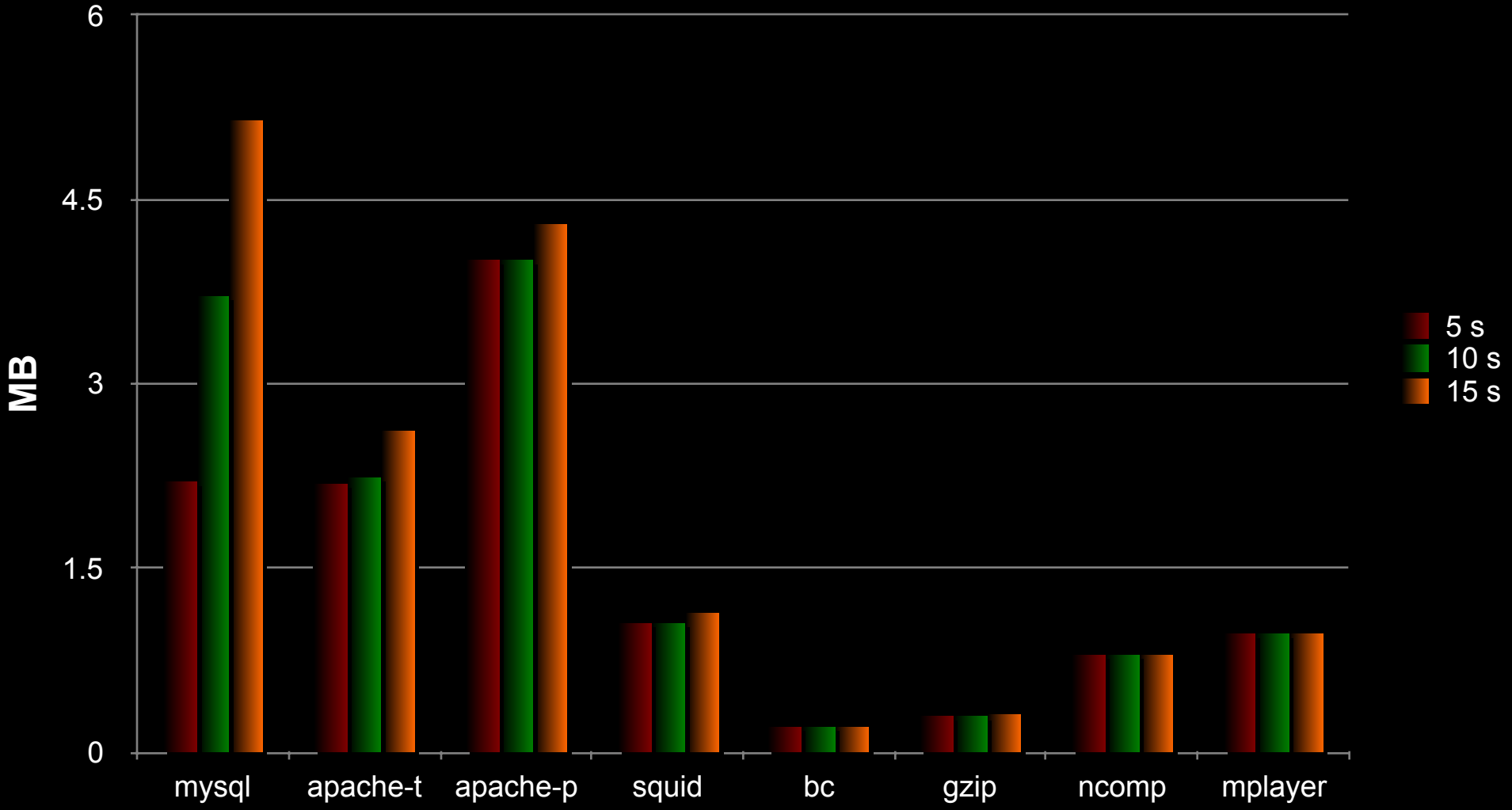
- Record-Replay on Debian: IBM HS-22 blade center
- Replay on Gentoo: VMware Fusion on MacBook Pro, different Linux kernel
- Replay on Windows: Lenovo T61p

Runtime Performance



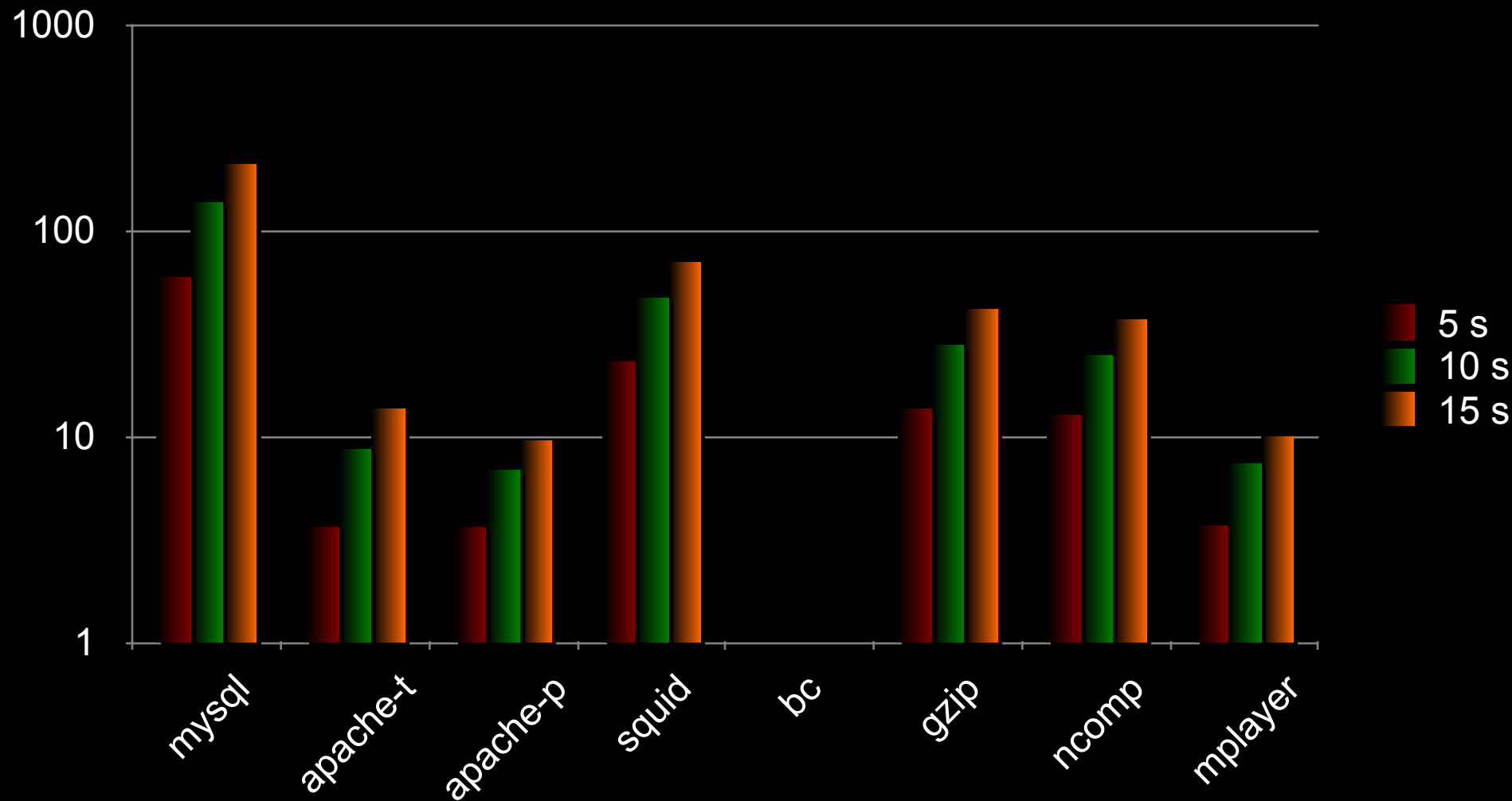
- Overhead below 3% for all except Squid (9%) and MySQL (17%)
- Replay is faster than recording

Partial Checkpoint Size



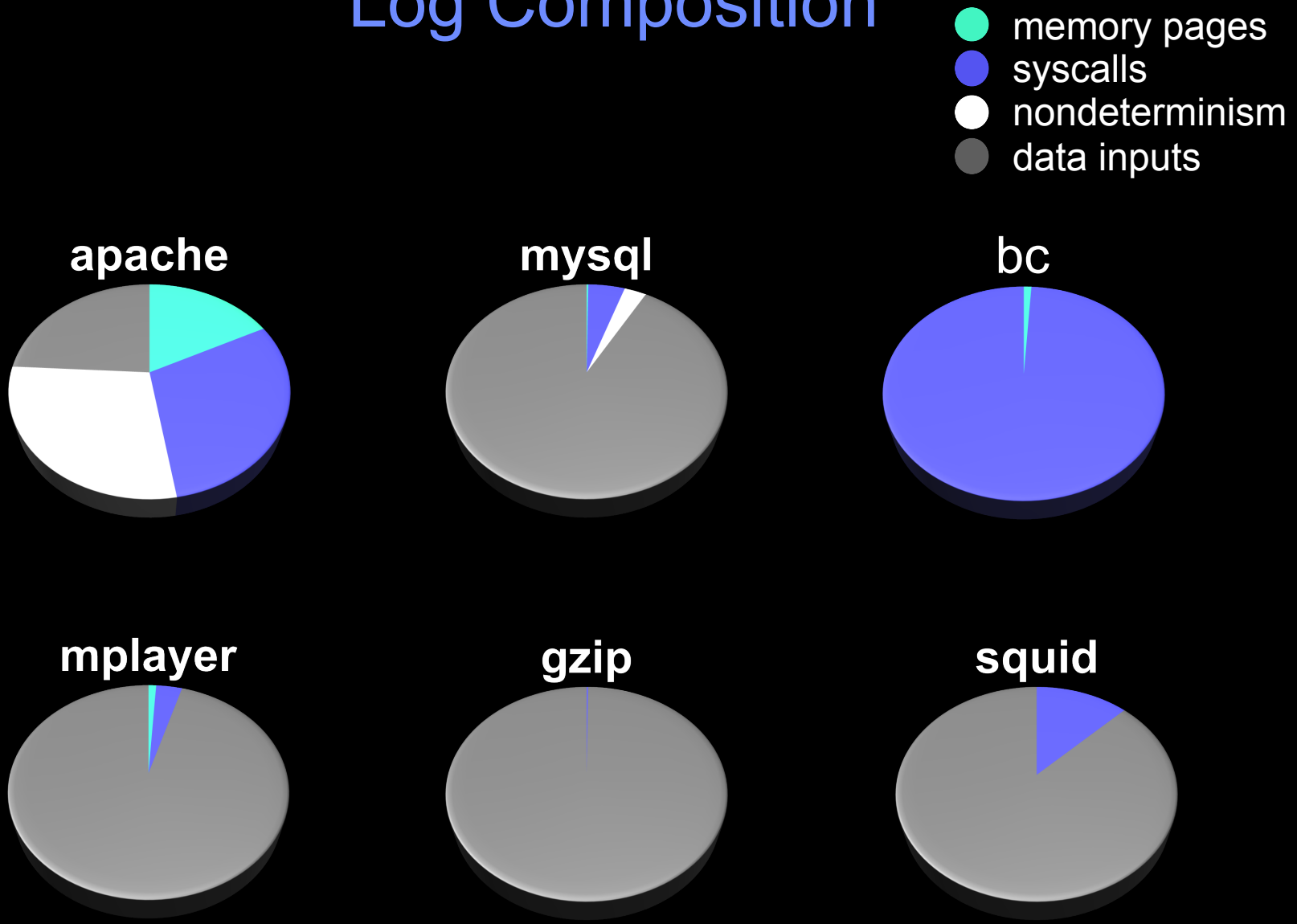
Largest partial checkpoint was ~5 MB -- fraction of application memory footprint

Log Size



- Grows with recording interval but not with application run time
- Largest log size for a 5 s interval was 59 MB

Log Composition



dinesh@apporbit.com